

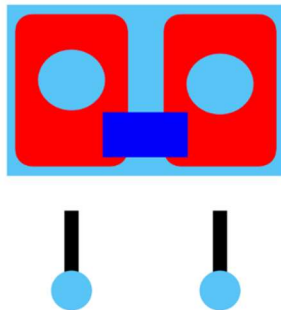
MyDesign Portfolio

Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

Design Idea #1

Description: Two short-range launching systems are designed to throw voting objects into two attached containers.



Drawing that includes distinguishing information such as scale, dimensions, and materials: The total footprint of the design is 10” in length and 9” wide. The launching system is 3D printed, while the rest of the design is either 3D printed or integrated plastic structures.

Design Idea #2

Description: The design has the same basic structure as the picture depicted below, but with some modifications. The extraneous structures like the plinko board at the bottom would be removed, the jars would be enlarged and attached more securely to the structure. The containers holding the voting objects would also incorporate designs for removing voting objects at the end of the day, like a release mechanism that allows the entire container to be lifted out from the structure and emptied.



The dimensions of the design are 11 inches in length, 8 inches in depth, and 12 inches in height. The structure would be 3D printed.

Design Idea #3

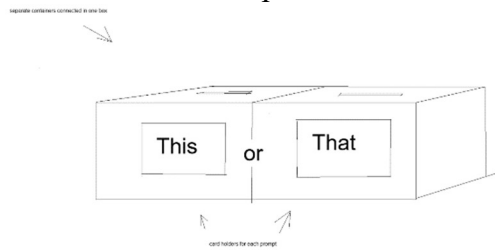
The basic structure is similar to the object depicted below. However, the containers would not be transparent and have wide open tops, allowing people to view the voting objects in both containers. The containers would also be detachable.



The structure would be 3D printed, its dimensions are 10 inches in length, 8 inches in depth, and 10 inches in height

Design Idea #4

Description: Two side by side buckets fashioned like cubes. The cubes are attached to one another and both secured with command strips. The front of the box will have the prompt card on them. Students will place tokens in each container.

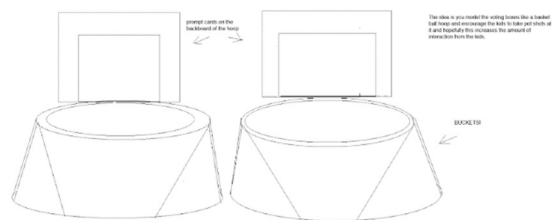


The design is 3D printed

Design Idea #5

Description: Two side by side buckets fashioned like basketball hoops. The hoops are attached to one another and both secured with command strips. The back board of the hoop will have the prompt card on them. Students will try to make a shot into the hoop with a pompom and that's how they will vote for the This or That game.

Drawing

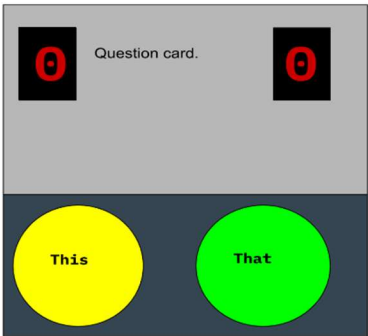


Dimensions: The total measurements of the design will be **10in 8in 5in** (x,y,z) each hoop will be **5in 8in 5in** (x,y,z) with a circumference of **15.7in**. The design is 3D printed.

Design Idea #6

Description: There are two different screens showing how many times the two options have been chosen. The options are chosen not through voting objects but by pressing a button

Drawing that includes distinguishing information such as scale, dimensions, and materials:



This design is 10 inches tall and several inches deep. The structure and buttons would be 3D printed and contain the electronic systems.

Design Idea Comparison and Selection (D1, D2)

	weight	Design1 (Launching Device)	Design2 (pachinko device)	Design3 (see though device)	Design 4 (Boxes)	Design 5(Basketball- like hoops)	Design 6 (Buttons
Size	0.3	1.2	1.2	1.2	1.2	1.2	1.2
Durability	0.2	-0.8	0.6	0.4	0.2	0.6	0.8

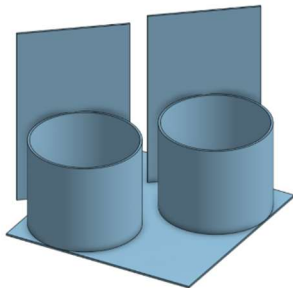
Safety	0.2	0.4	0.4	0.8	0.6	0.6	0.6
Fun Factor	0.1	0.4	-0.1	-0.2	-0.4	0.2	0.4
Cost	0.1	0.4	0.1	0.3	0.3	0.4	0
Complexity	0.1	0	0.2	-0.1	0.2	0.3	0.2
score	Totals	1.6	2.4	2.4	2.1	3.3	3.2

Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

The design (DM 5) prototype would be made out of cardboard or 3D printed, which has a very low cost.

Design Blueprint #1



Manufacturing Methods:

	Details	Tools and Manufacturing Methods used
Step 1	Cut cardboard into the pieces needed	Cardboard , Box cutter ,(Cutting cardboard). Measurements and cut out four pieces of 7.85in by 5in, one piece of 10in by 5in, and two pieces of 4in by 8in. 8 pieces 4in by 2 in
Step 2	Bend the cardboard	Bend the four pieces of 7.85 by 5 short ways you should have four crescent moon shaped pieces.

		Use a boxcutter to cut vertical slits as needed while bending the cardboard to aid in the structural stability of the cardboard.
Step 3	Tape/Glue the pieces together.	Using the 10" by 5" piece of cardboard as a base on one half of the sheet, glue the two crescent moon shaped pieces together then secure them to the sheet with glue as well. Repeat on the other side of the base piece with the additional crescent moon shaped pieces to create a hoop. Wrap tape around the hoop so secure.
Step 4	Add support	Bend all of the 4in by 2in pieces into L shapes and glue on the inside of the hoop along the seams of the crescent moon in an X shape.
Step 5	Tape/Glue the pieces together.	With pieces with less contact area tape them together and pieces with high contact area glue them together like the backboard.

Supply List and Costs:

Item	Cost each	# needed	Total
cardboard	0	3-5	0
scissors	\$4	1	\$4
tape	\$7	1	\$7
glue	\$1	1	\$1
Box cutter	\$6	1	\$6

Total Cost			\$18
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Plant of Action

Gantt Chart

Task	Details	Week 1	Week 2	Week 3	Week 4	Week 5
Define the Problem	Identify the need for a “This or That” voting game for a middle school library. Decide the main goal, users, and setting.					
Research Prior Designs	Look at examples of library engagement activities, voting systems, and passive programming ideas.					
Create Design Requirements	List important requirements such as size, durability, safety, cost, visibility, and ease of use.					
Brainstorm Design Ideas	Create multiple possible designs, including containers, hoops, buttons, and other voting systems.					
Compare Designs	Use a decision matrix to compare each idea based on size, durability, safety, fun factor, cost, and complexity.					
Select Final Design	Choose the basketball-style hoop design because it is fun, durable, simple, and easy for students to understand.					
Create Blueprint and CAD Sketches	Make sketches, CAD drawings, dimensions, supply list, and building instructions so the design can be built clearly.					

Gather Materials	Collect cardboard, tape, glue, scissors, box cutter, and pom poms.		
Build Prototype	Cut, bend, tape, and glue the cardboard pieces together to create the base, hoops, walls, and supports.		
Test Prototype	Run tests for size, durability, shaking, flipping, throwing accuracy, and container capacity.		
Analyze Data	Review test results and explain what the data shows about durability, size, capacity, and ease of use.		
Collect Feedback	Ask stakeholders, users, and field experts for feedback on what works and what should improve.		
Final Improvements	Use test results and feedback to suggest improvements like better pom pom storage, bigger targets, and easier resetting.		